Image Subset

Image Subset

- **Because of the very large size of many remotely sensed images, analysts typically work with those segments of full images that specifically pertain to the task at hand**
- **Could be one of the first tasks in each project is to prepare *subsets*, portions of larger images selected to show only the region of interest**

Image Subset

- **Working with the subset image helps to**
  - **Minimize computer storage and processing**
  - **Save the analyst's time and effort**

Image Subset

- **If the subset is too small, then it may be difficult to identify sufficient landmarks**
- **Therefore, it may be useful to prepare a preliminary subset large enough to conduct the image processing, a smaller subset for analytical use**
- **Subsets should be large enough to provide the context required for the specific analysis at hand**

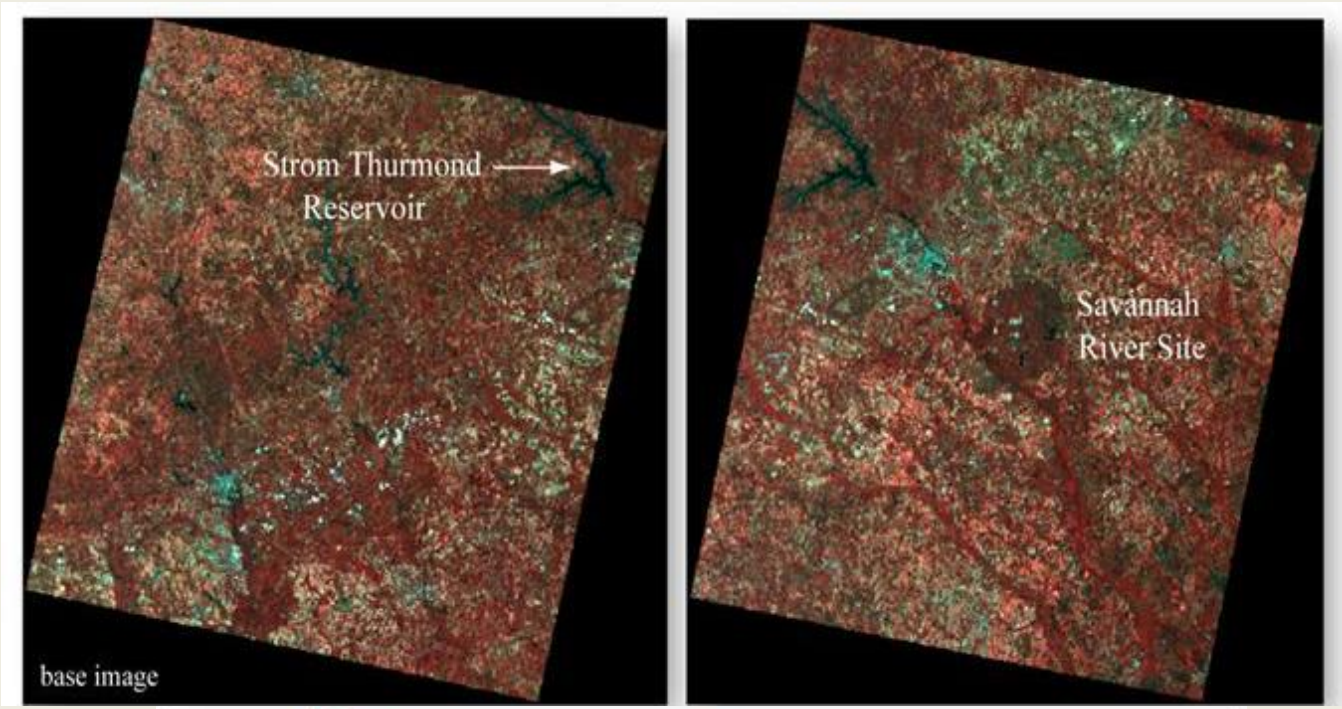
Image Mosaicking

Image Mosaicking

- The process of seamlessly stitching together or blending a set of overlapping images of an area into one large image
- It is used in
  - Map making using several geo-rectified images
  - Tracking image changes
  - Creating panoramic images by integrating a sequence of overlapping images
  - 3-D scene construction by integrating multiple view images

Image Mosaicking

- Mosaicking several rectified images requires several steps:
  - Individual images should be rectified to the same map projection and datum
  - Ideally, rectification of the *n* images is performed using
    - Same pixel size
  - One of the images to be mosaicked is designated as the *base image*.
    - The base image and *image 2* should normally overlap a certain amount (e.g., 20% to 30%)
- The focus should be on finding the best seam between the images in the overlapping area


Strom Thurmond
Reservoir
base image
Savannah
River Site

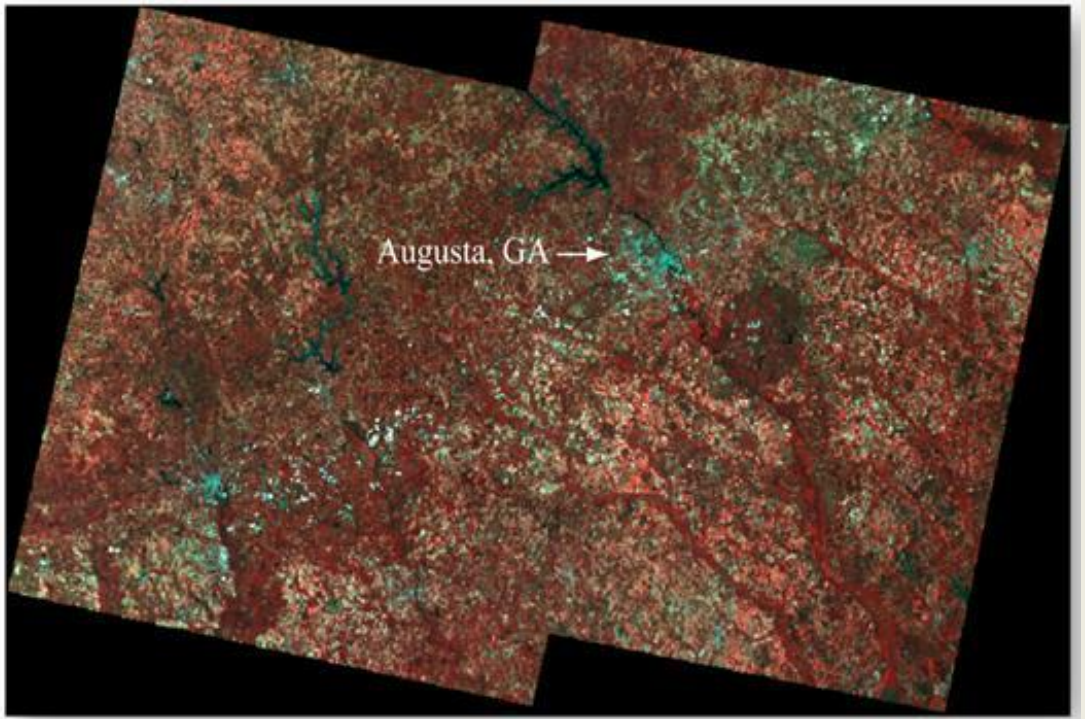
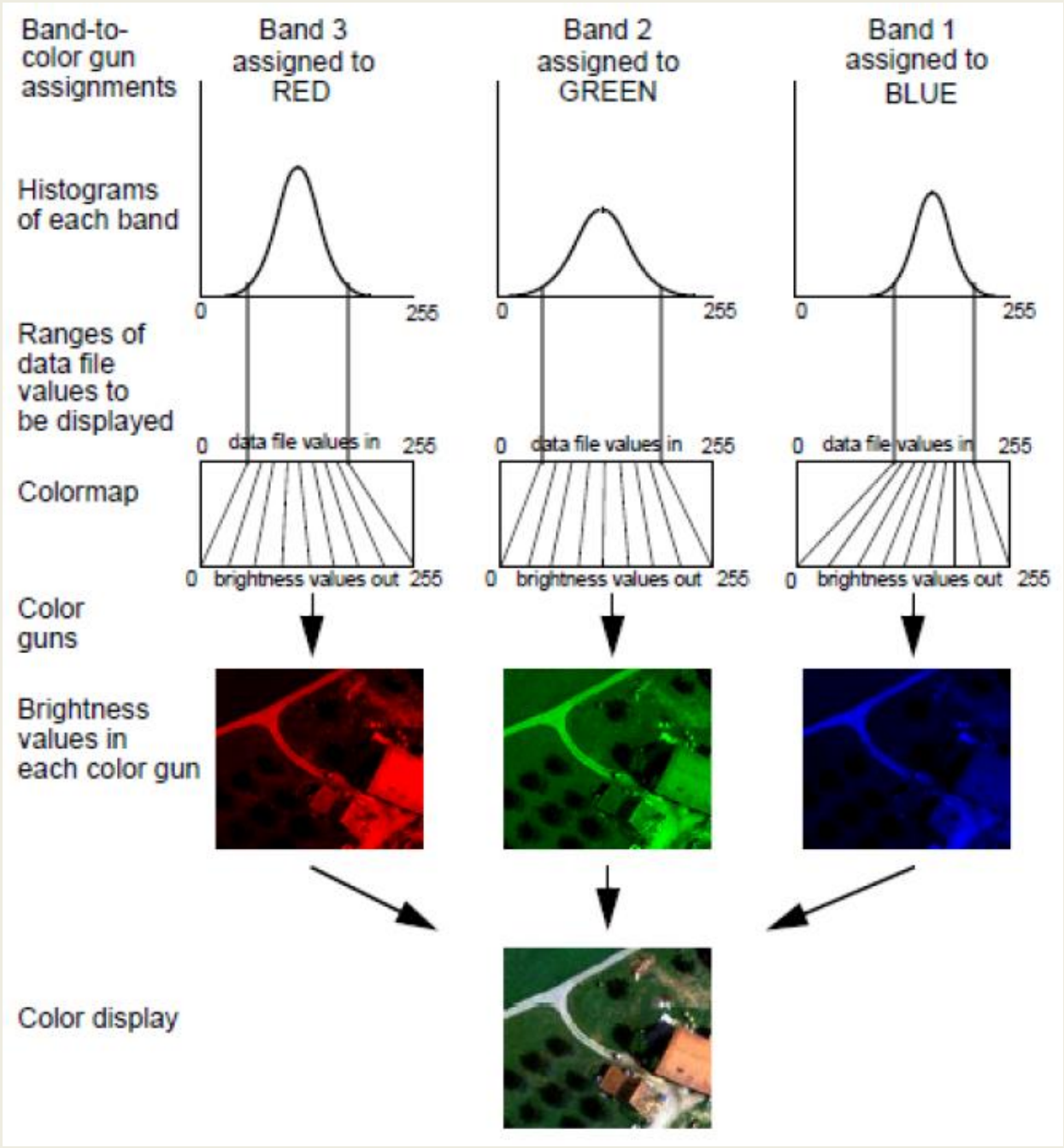

Augusta, GA

Image Display

Image Display

- **A natural-color image approximates the colors that would appear to a human observer of the scene**

- **A color-infrared image shows the scene as it would appear on color infrared film, which is familiar to many analysts**


Band-to-color gun assignments
Band 3 assigned to RED
Band 2 assigned to GREEN
Band 1 assigned to BLUE
Histograms of each band
0
255
Ranges of data file values to be displayed
data file values in
Colormap
brightness values out
Color guns
Brightness values in each color gun
Color display

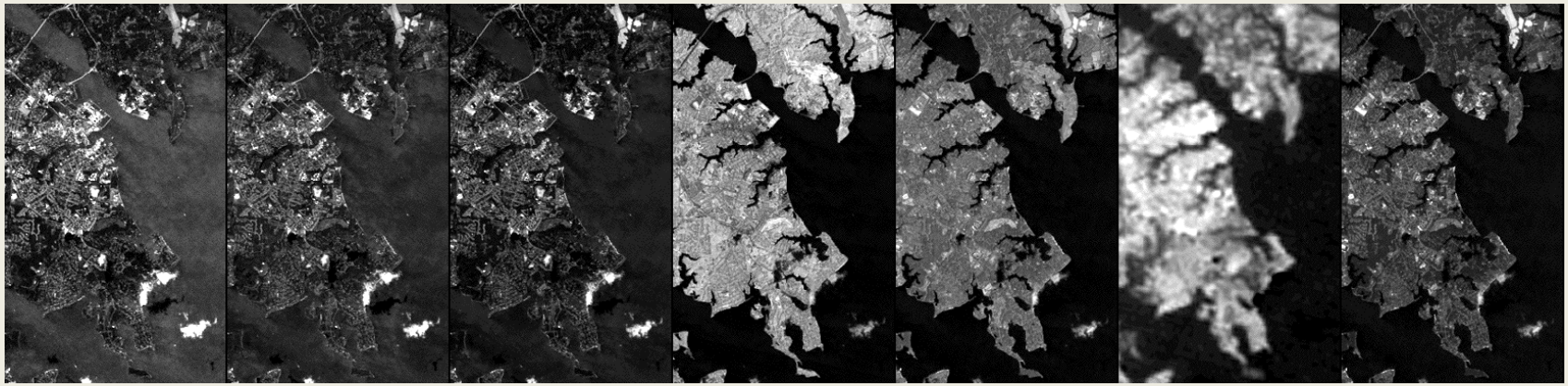
Different kinds of image

- Panchromatic image
- True-color image
- False-color image









Visible

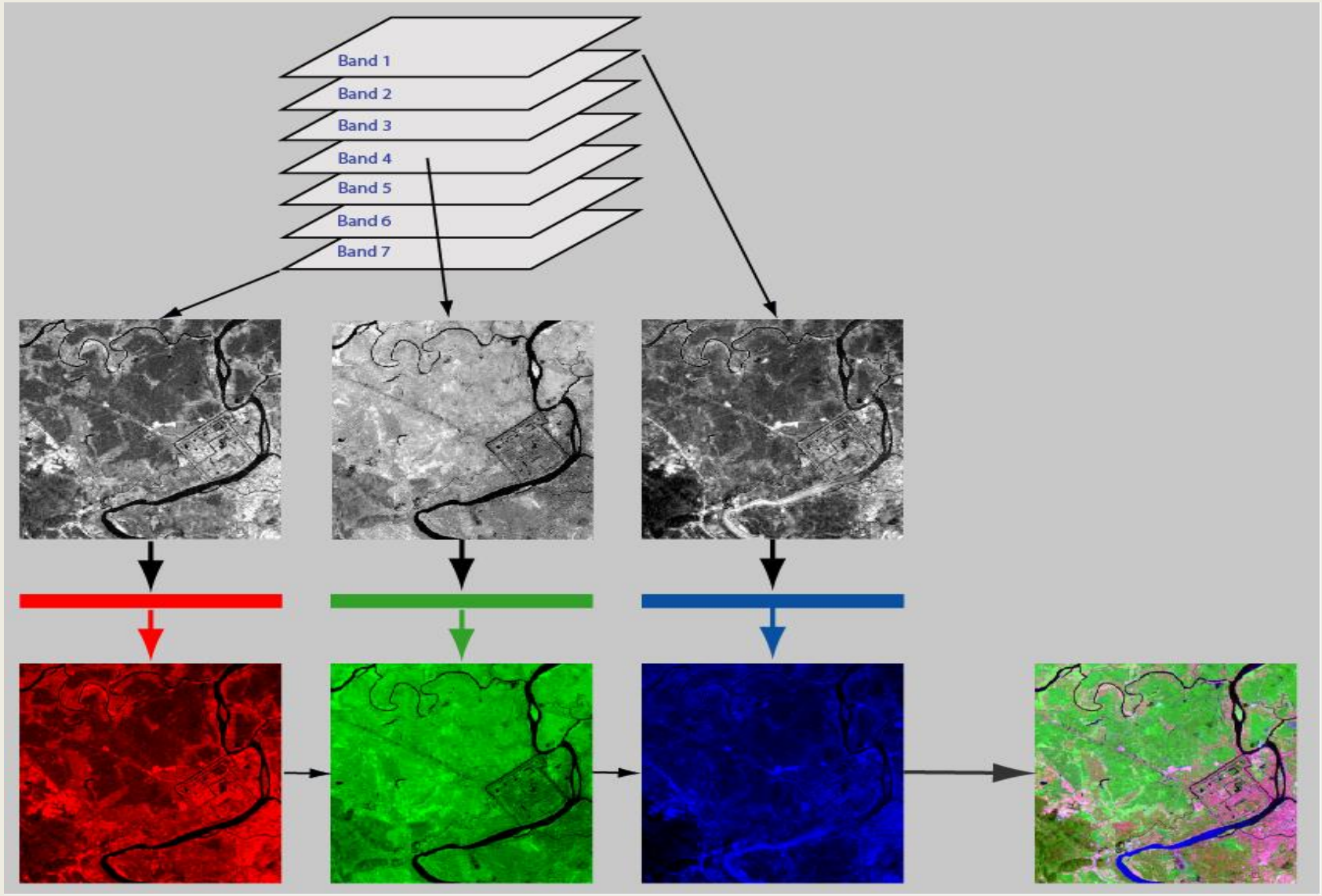
Infrared

1 2 3 4 5 6 7

3,2,1



RGB Band Composite



Landsat ETM+ bands 3,2,1 – Penetrates shallow water and shows submerged shelf, water turbidity



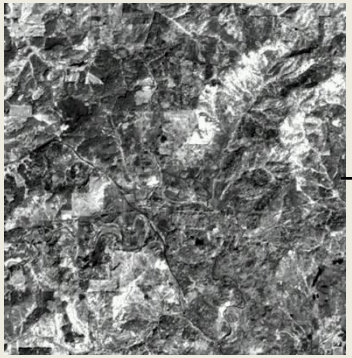
Landsat ETM+ bands 4,3,2 – Peak chlorophyll, land/water boundary, urban areas



Near Infra Red Composite
RGB = 4,3,2

- Blue visible band is not used
- Visible green sensor band to the blue color gun
- Visible red sensor band to the green color gun
- NIR band to the red color gun
- Results in the familiar NIR composite with vegetation portrayed in red

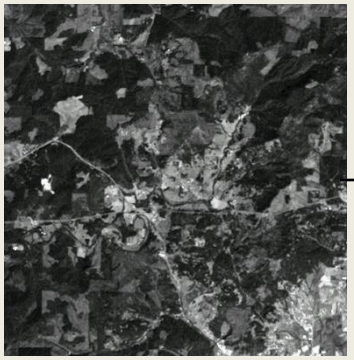
Digital Image Display



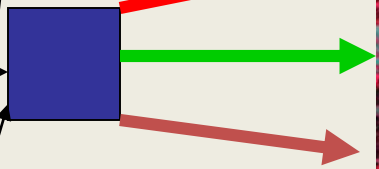
Band 4 (0.7-0.9 μm)

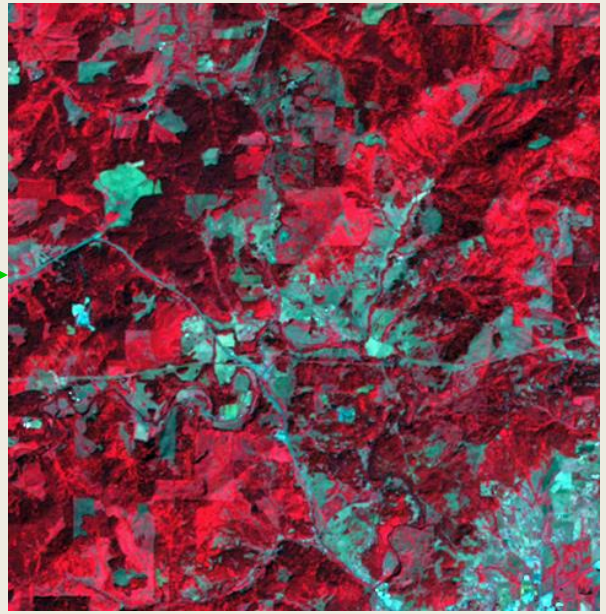


Band 3 (0.55-0.7 μm)



Band 2 (0.45-0.55 μm)

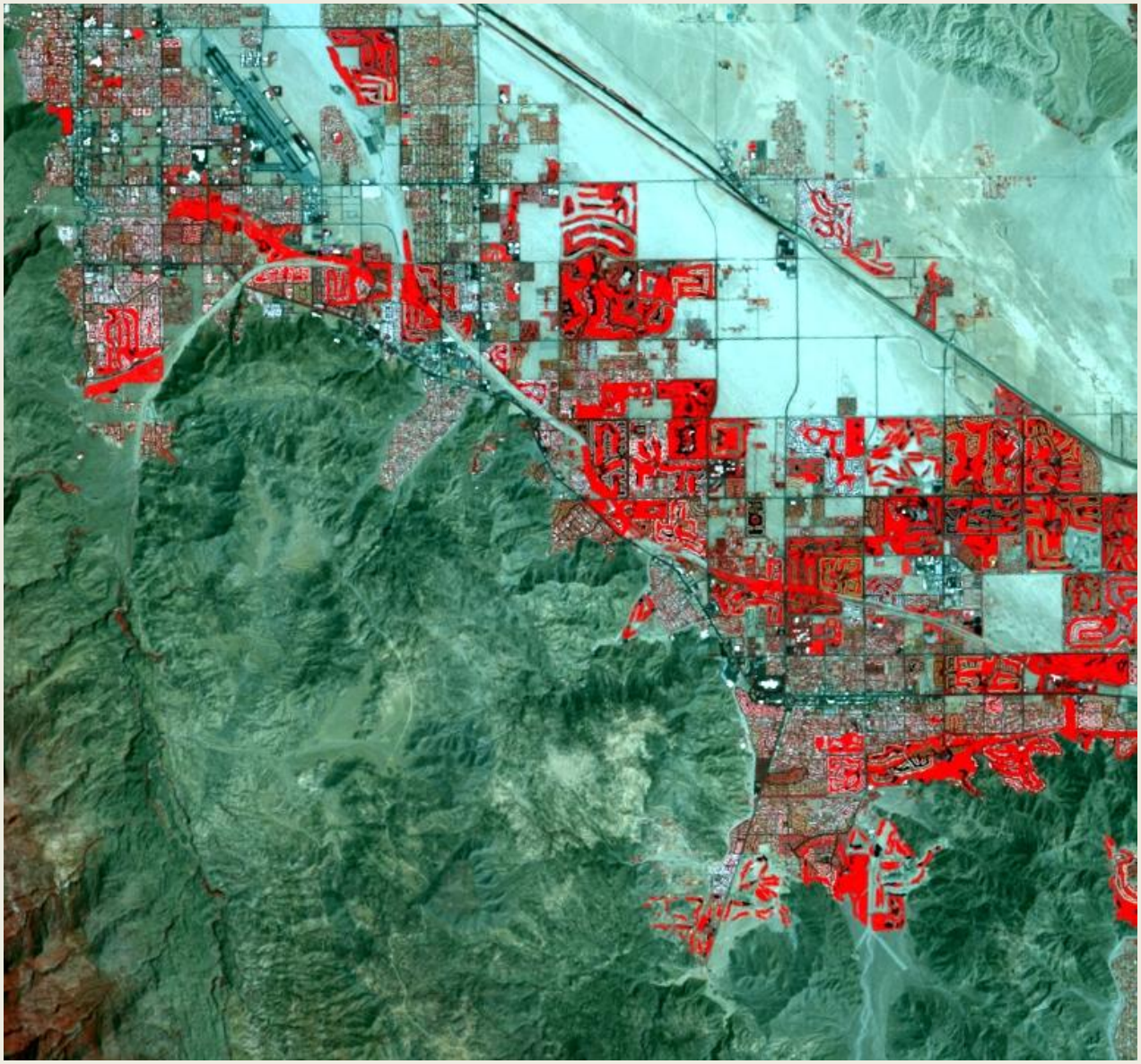




RGB:432 (False Color Composite)

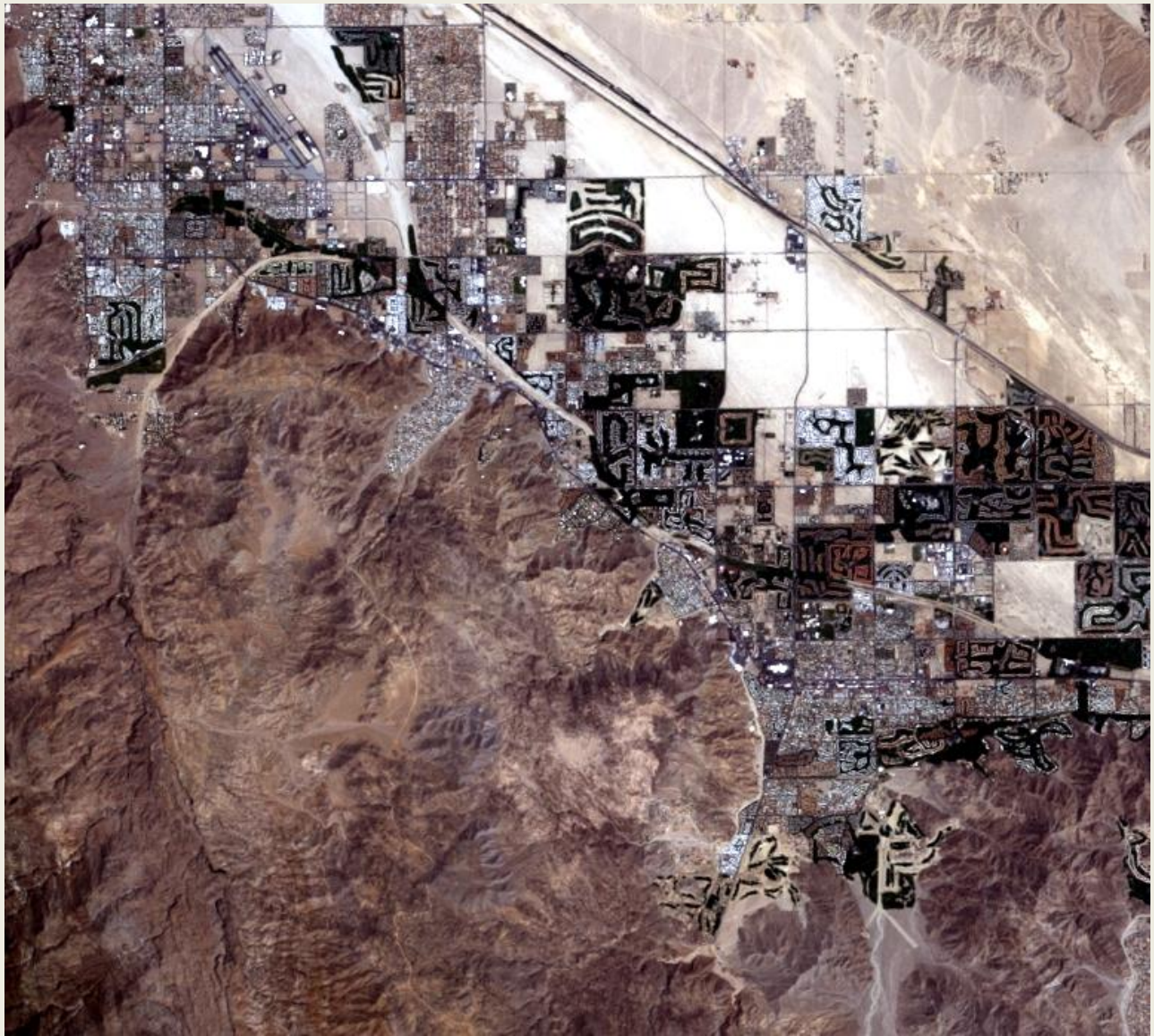
Palm Springs, CA

Bands 4, 3, 2 (NIR, red, green)



Palm Springs, CA

Bands 3,2,1 (red, green, blue)



Depending upon the band combination and colors assigned, land cover appears in various colors

	True Color Red: Band 3 Green: Band 2 Blue: Band 1	False Color Red: Band 4 Green: Band 3 Blue: Band 2	SWIR (GeoCover) Red: Band 7 Green: Band 4 Blue: Band 2
Trees and bushes	Olive Green	Red	Shades of green
Crops	Medium to light green	Pink to red	Shades of green
Wetland Vegetation	Dark green to black	Dark red	Shades of green
Water	Shades of blue and green	Shades of blue	Black to dark blue
Urban areas	White to light blue	Blue to gray	Lavender
Bare soil	White to light gray	Blue to gray	Magenta, Lavender, or pale pink

